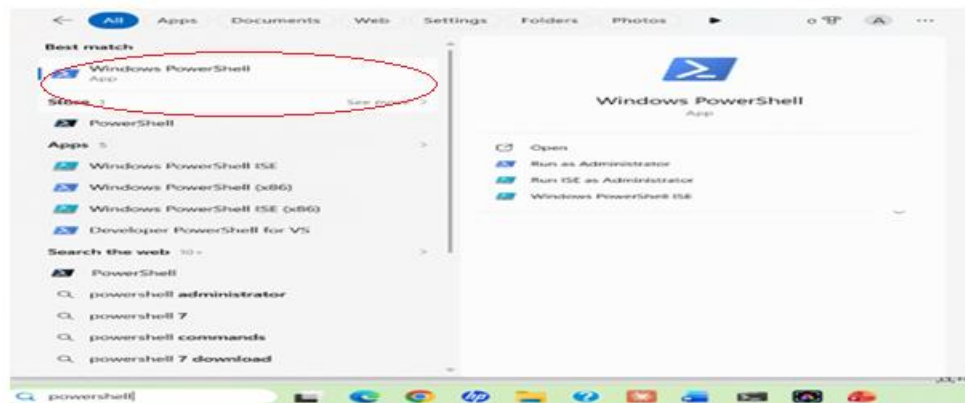


Name:Monisha Raghini Chelle

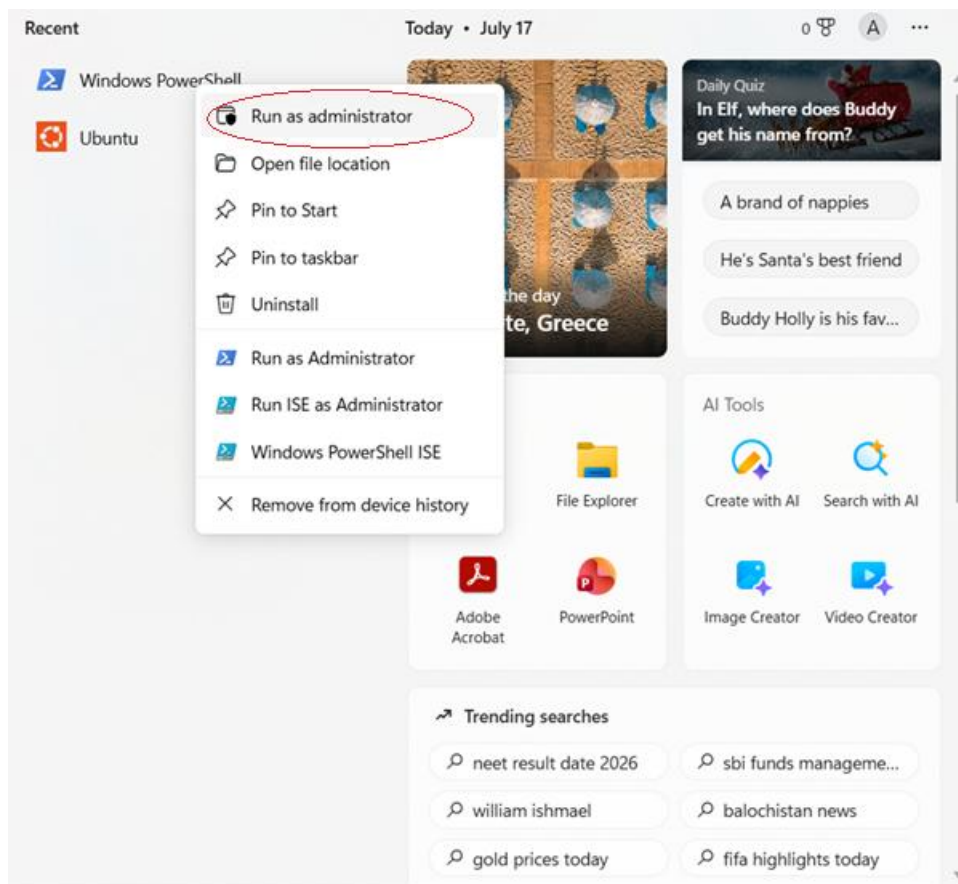
Roll No:2520030537

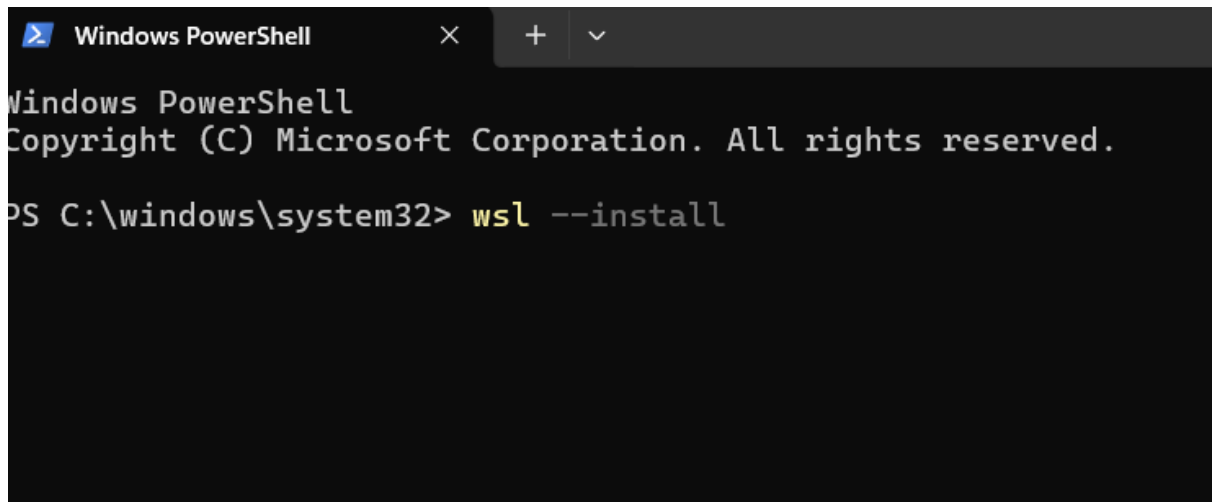
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', and 'PS C:\windows\system32> wsl --install'. The 'wsl' command is highlighted in yellow.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

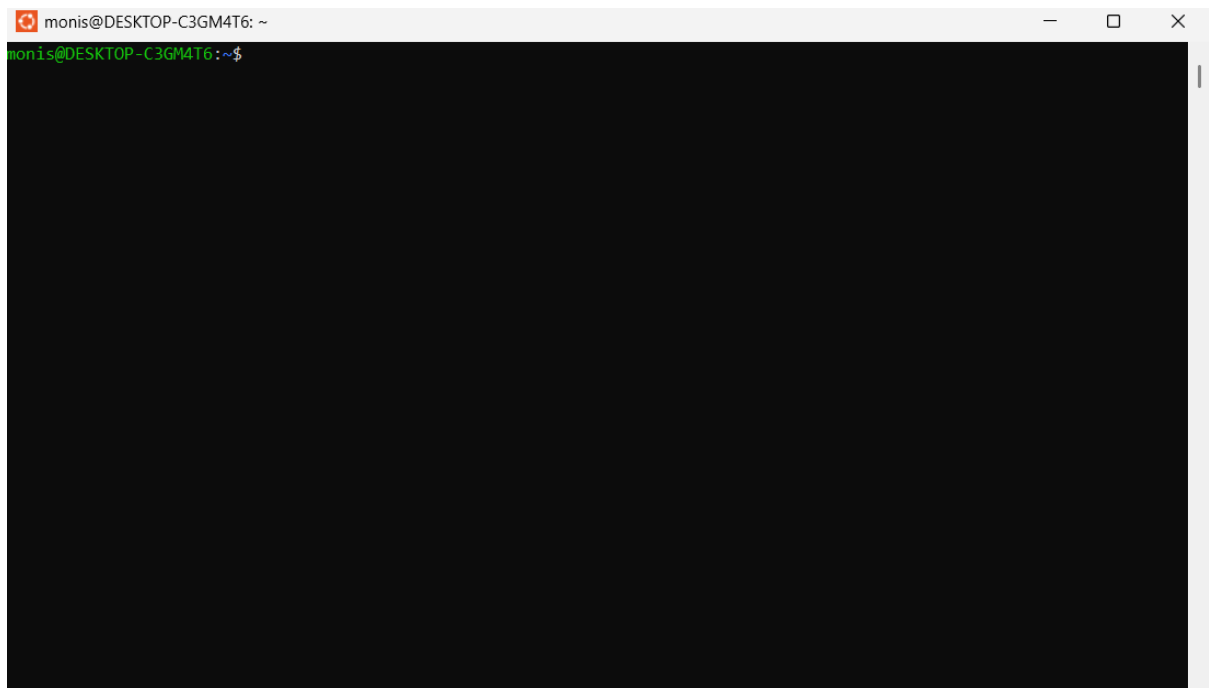
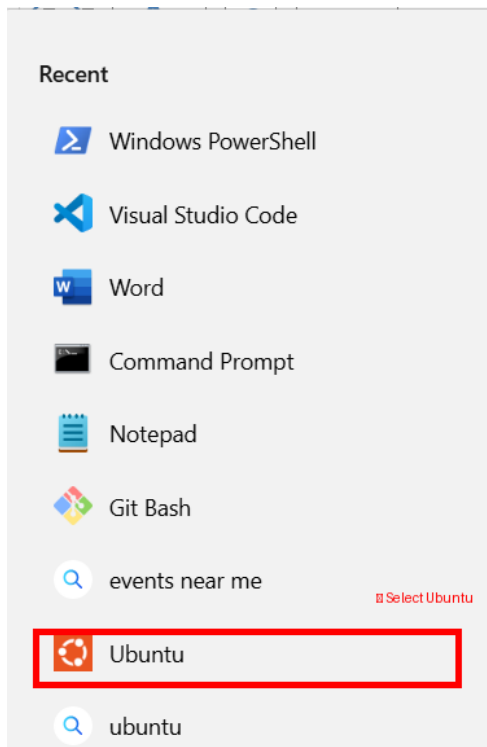
Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****



Step 6: Update Ubuntu

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ sudo apt update  
[sudo: authenticate] Password:  
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease  
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]  
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]  
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [422 kB]  
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.6 kB]  
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]  
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]  
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]  
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]  
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [357 kB]  
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [69.0 kB]  
Hit:12 http://archive.ubuntu.com/ubuntu resolute-backports InRelease  
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [495 kB]  
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]  
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]  
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [250 kB]  
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [83.2 kB]  
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]  
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [371 kB]  
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [71.7 kB]  
Fetched 3190 kB in 1s (2455 kB/s)  
124 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC (C Compiler)

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ sudo apt install build-essential  
Upgrading:  
  build-essential  
  
Installing dependencies:  
  libcrypt-dev  
  
Summary:  
  Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 123  
  Download size: 0 B / 127 kB  
  Space needed: 384 kB / 1024 GB available  
  
Continue? [Y/n] y  
Selecting previously unselected package libcrypt-dev:amd64.  
(Reading database ... 42277 files and directories currently installed.)  
Preparing to unpack .../libcrypt-dev_1%3a4.5.1-1_amd64.deb ...  
Unpacking libcrypt-dev:amd64 (1:4.5.1-1) ...  
Preparing to unpack .../build-essential_12.12ubuntu2.26.04.2_amd64.deb ...  
Unpacking build-essential (12.12ubuntu2.26.04.2) over (12.12ubuntu2) ...  
Setting up libcrypt-dev:amd64 (1:4.5.1-1) ...  
Setting up build-essential (12.12ubuntu2.26.04.2) ...  
Processing triggers for man-db (2.13.1-1build1) ...  
monis@DESKTOP-C3GM4T6:~$
```

Step 8: Verify the Installation

```
monis@DESKTOP-C3GM4T6: ~  
monis@DESKTOP-C3GM4T6:~$ gcc --version  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
monis@DESKTOP-C3GM4T6:~$
```

Step 9: Create Your C Program

Code :

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main(void)
```

```
{
```

```
    char command[100];
```

```
    printf("Enter a Linux command: ");
```

```
    scanf("%99s", command);
```

```
    pid_t pid = fork();
```

```
    if (pid < 0)
```

```
    {
```

```
        perror("Fork failed");
```

```
        return 1;
```

```
    }
```

```
    if (pid > 0)
```

```
    {
```

```
// Parent Process

printf("\nParent Process\n");

printf("Parent PID : %d\n", getpid());

printf("Child PID : %d\n", pid);


wait(NULL);


printf("\nChild process completed.\n");
}
else
{
    // Child Process

    printf("\nChild Process\n");

    printf("Child PID : %d\n", getpid());

    printf("Parent PID: %d\n", getppid());


    execlp(command, command, (char *)NULL);


    // Executes only if execlp() fails

    perror("Command execution failed");

    exit(1);
}

return 0;
}
```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill

GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%99s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("Fork failed");
        return 1;
    }

    if (pid > 0)
    {
        // Parent Process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }
    else
    {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, (char *)NULL);

        // Executes only if execlp() fails
        perror("Command execution failed");
        exit(1);
    }
}
```

 Help
 Exit

 Write Out
 Read File

 Where Is
 Replace

 Cut
 Paste

 Execute
 Justify

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1
#include <sys/wait.h>

int main(void)
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%99s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("Fork failed");
        return 1;
    }

    if (pid > 0)
    {
        // Parent Process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }
    else
    {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, (char *)NULL);

        // Executes only if execlp() fails
        perror("Command execution failed");
        exit(1);
    }

    return 0;
}

Write to File: process.c
^G Help      M-D DOS Format
^C Cancel    M-M Mac Format
```

Example: nano process.c


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  Program2      Program4.c  Simple-Commands-Week1  program2      program5.c
OSLAB                          Program2.c    Program5     example.txt             program2.c    sample1
OSLAB14                        Program3      Program5.c  foldername              program3      sample1.c
OSSP-Practical                 Program3.c    Program6     hello                   program3.c    sample1.c.s
ave
Program1                       Program4      Program6.c  hello.c                 program4
Program1.c                     Program4.1    Program7     moni                    program4.c
Program1.c.save                Program4.1.c  Program7.c  monis                   program5
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ mkdir skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano process.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc process.c -o process
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./process
Enter a Linux command: pwd

Parent Process
Parent PID : 1323
Child PID  : 1324

Child Process
Child PID : 1324
Parent PID: 1323
/home/monis/OSLAB/skill

Child process completed.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 10: Compile the Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc process.c -o process
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./process
Enter a Linux command: pwd

Parent Process
Parent PID : 1323
Child PID  : 1324

Child Process
Child PID : 1324
Parent PID: 1323
/home/monis/OSLAB/skill

Child process completed.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

 2520030537-OSSP

Public

Pin

 main

 1 Branch

 0 Tags

Go to file

Add file

<> Code

 monishachelle-max	Added OSSP Practical 3 and 4	477e86d · last week	16 Commits
	2520030537_Practical	Added OSSP Practical 3 and 4	last week
	2520030537_Skill	Added 2520030537_Skill folder	3 weeks ago

README



Add a README

Help people interested in this repository understand your project.

Add a README